

# Structure Information Graph Learning with Graph Convolutional Networks for Anomaly Detection in Mixed-Attribute Data

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**Abstract**—Anomaly detection in mixed-attribute data remains a challenging problem due to heterogeneous feature types and the difficulty of modeling structural relationships among samples. Traditional distance-based and neighborhood-based methods often struggle with the complex interactions between numerical and categorical attributes, while recent graph neural network approaches rely heavily on predefined graphs that may not capture intrinsic data structure. In this paper, we propose NMIGOD, a novel anomaly detection framework that integrates Neighborhood Mutual Information (NMI) based graph construction with Graph Convolutional Networks (GCN) for semi-supervised anomaly detection in mixed-attribute data. First, an entropy-based adaptive radius mechanism is introduced to determine per-attribute neighborhood scales without manual tuning. Second, a structure information graph is constructed using NMI to capture structural similarity between objects across heterogeneous attribute spaces. Finally, a two-layer GCN is trained in a semi-supervised manner on the learned graph to produce anomaly scores. Extensive experiments on 30 real-world UCI datasets demonstrate that NMIGOD significantly outperforms five state-of-the-art methods, achieving an average AUC of 0.8823 and the highest F1-score of 0.5542. Ablation studies and statistical tests confirm the contribution of each core component and the significance of the improvements.

**Index Terms**—Anomaly detection, neighborhood mutual information, graph structure learning, graph convolutional networks, mixed-attribute data.

## I. INTRODUCTION

**A**NOMALY detection aims to identify rare patterns or observations that deviate markedly from the majority of data [1], [2]. Since anomalies often signal critical events, it has become a fundamental task in numerous real-world applications, including financial fraud detection [3], cybersecurity intrusion detection [4], and industrial fault diagnosis.

Existing anomaly detection methods can be broadly divided into three categories. The first category comprises traditional approaches, including statistical models [5], distance-based methods such as  $k$ -nearest neighbors [6], density-based methods such as LOF [7], and cluster-based methods [8]. While

these methods are well-studied and computationally efficient, they often struggle with high-dimensional data and heterogeneous feature types [9].

The second category consists of deep learning-based methods, including autoencoder-based approaches [10] and graph neural network methods. GNN-based methods [22]–[24] have shown promising results by leveraging relational information among samples. However, most GNN-based anomaly detection methods rely on predefined graphs (e.g.,  $k$ -NN graphs) that are constructed independently of the downstream detection task, limiting their ability to capture task-relevant structure.

The third category encompasses rough set and formal concept analysis (FCA) based approaches. Neighborhood rough set (NRS) methods [11]–[13] effectively handle heterogeneous data through neighborhood granulation. The NIEOD method [14] uses neighborhood information entropy with HEOM distance for anomaly scoring. ADFNR [16] employs adaptive density fuzzy neighborhood rough sets, and DASOD [20] leverages dual-aspect formal concept analysis for outlier detection with structural deviation and attribute rarity.

As stated above, both NRS-based and GCN-based approaches have complementary strengths and weaknesses, making their integration a natural strategy. For achieving this, the key issue is how to construct a meaningful graph that faithfully captures the structural relationships inherent in mixed-attribute data. To address this challenge, we propose the Neighborhood Mutual Information (NMI) based graph and the NMIGOD framework. The main contributions of this paper are summarized as follows:

- 1) An entropy-based adaptive radius is proposed to determine neighborhood scales without manual tuning;
- 2) A structure information graph is constructed using Neighborhood Mutual Information to capture structural similarity;
- 3) A novel anomaly detection framework named NMIGOD is proposed by integrating NMI-based graph construction with GCN for semi-supervised learning.

The remainder of this paper is organized as follows: Section 2 reviews the preliminaries. Section 3 describes the proposed NMIGOD framework in detail. Section 4 reports the experimental results, and Section 5 concludes the paper.

## II. PRELIMINARIES

This section reviews two essential preliminaries: neighborhood information systems and graph convolutional networks.

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### A. Neighborhood Information System

A neighborhood information system is an extension of the classical information system by incorporating neighborhood relations. An information system is formally defined as a quadruple  $\mathcal{I} = (U, A, V, f)$ , where  $U$  is a non-empty finite set of objects,  $A$  is a non-empty finite set of attributes,  $V = \bigcup_{a \in A} V_a$  is the union of attribute domains, and  $f : U \times A \rightarrow V$  denotes an information function such that  $f(x, a) \in V_a$  for every  $x \in U, a \in A$ .

The construction of neighborhoods often relies on a distance function defined over the attribute set, which serves as the basis for granulating the universe and establishing neighborhood relations.

**Definition 1** [14]. Let  $\mathcal{I} = (U, A, V, f)$  be a neighborhood information system. For  $B \subseteq A$ , an attribute subset, and a radius  $\delta$ , the neighborhood of object  $x$  with respect to  $B$  is defined as:

$$n_B^\delta(x) = \{y \in U \mid \Delta_B(x, y) \leq \delta\} \quad (1)$$

where  $\Delta_B$  is a distance function. That is,  $n_B^\delta(x)$  collects all objects whose distance from  $x$  on the attribute subset  $B$  does not exceed  $\delta$ .

The neighborhood relation induced on  $U$  by  $B$  is defined as:

$$R_B^\delta = \{(x, y) \in U \times U \mid y \in n_B^\delta(x)\} \quad (2)$$

This relation is reflexive and symmetric. A neighborhood cover of  $U$  is constituted by the quotient set  $U/R_B^\delta$ , forming the underlying granular structure for subsequent analysis.

A neighborhood information system (NIS) is then defined as  $\mathcal{N} = (U, A, V, f, R_A^\delta)$ , where  $R_A^\delta$  denotes the neighborhood relation induced by the full attribute set  $A$ . When a decision attribute  $D$  is present, the system extends to a neighborhood decision system  $\mathcal{N}_D = (U, A \cup \{D\}, V, f, R_A^\delta)$ .

### B. Graph Convolutional Networks

Graph Convolutional Networks (GCNs) [21] extend the convolution operation from image data to graph-structured data. Given a graph with adjacency matrix  $\mathbf{A}$  and node feature matrix  $\mathbf{X}$ , each GCN layer updates node representations through the following propagation rule:

$$\mathbf{H}^{(l+1)} = \sigma \left( \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)} \right) \quad (3)$$

where  $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$  denotes the adjacency matrix with added self-loops,  $\tilde{\mathbf{D}}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij}$  is the corresponding degree matrix,  $\mathbf{H}^{(l)}$  are the node representations at layer  $l$  (with  $\mathbf{H}^{(0)} = \mathbf{X}$ ), and  $\mathbf{W}^{(l)}$  is a trainable weight matrix.

For a two-layer GCN, the forward propagation is:

$$\mathbf{Z} = \text{softmax} \left( \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \text{ReLU} \left( \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{X} \mathbf{W}^{(0)} \right) \mathbf{W}^{(1)} \right) \quad (4)$$

where  $\mathbf{Z} \in \mathbb{R}^{n \times 2}$  produces a probability distribution over the normal and anomalous classes.

In the semi-supervised setting, the model is trained with a masked cross-entropy loss that only accounts for labeled nodes:

$$\mathcal{L} = - \sum_{i \in \mathcal{V}_L} \sum_{c=0}^1 \mathbf{Y}_{ic} \ln \mathbf{Z}_{ic} \quad (5)$$

framework.pdf

Fig. 1. Overview of the proposed NMIGOD framework.

where  $\mathcal{V}_L$  denotes the set of labeled nodes,  $\mathbf{Y}_{ic}$  is the ground-truth label indicator (i.e.,  $\mathbf{Y}_{ic} = 1$  if node  $i$  belongs to class  $c$ , and otherwise 0), and  $\mathbf{Z}_{ic}$  is the predicted probability that node  $i$  belongs to class  $c$ .

## III. PROPOSED METHOD

Building upon the neighborhood information system and GCN introduced in Section 2, this section presents the proposed anomaly detection framework. The core idea is to construct a graph based on structural similarity measured by Neighborhood Mutual Information (NMI), and then perform GCN-based semi-supervised anomaly detection on the learned graph. The overall framework is illustrated in Fig. 1.

### A. Neighborhood System Construction

Given an information system  $\mathcal{I} = (U, A, V, f)$ , the construction of neighborhood knowledge requires a distance metric and a set of per-attribute thresholds. In Section 2.1, the classical neighborhood system is defined with a single global radius  $\delta$ . For heterogeneous data, we adopt the Heterogeneous Euclidean Overlap Metric (HEOM) [15] as the distance function and further propose an adaptive per-attribute radius mechanism.

For  $x, y \in U$ , the HEOM distance  $\Delta(x, y)$  with respect to  $A$  is defined as:

$$\Delta_A(x, y) = \sqrt{\sum_{a \in A} d_a(x_a, y_a)^2} \quad (6)$$

where  $d_a$  denotes the component-wise distance under attribute  $a$ , given by:

$$d_a(x_a, y_a) = \begin{cases} |x_a - y_a|, & \text{if } a \text{ is numerical} \\ \mathbf{1}[x_a \neq y_a], & \text{if } a \text{ is categorical} \end{cases} \quad (7)$$

**Algorithm 1** Adaptive Radius Computation

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**Input:** Data  $U$ , numerical attributes  $A_{num}$   
**Output:** Adaptive radii  $\{\varepsilon_a\}_{a \in A_{num}}$   
**for** each  $a \in A_{num}$  **do**  
    Compute  $\sigma_a \leftarrow \text{std}(x_a)$   
    Sort objects by  $x_a$ , compute gaps  $\Delta_i^a$   
    Split at gaps where  $\Delta_i^a > \sigma_a$ , form components  $\{C_1, \dots, C_k\}$   
    Compute  $NE_a \leftarrow -\sum_i \frac{|C_i|}{|U|} \log_2 \frac{|C_i|}{|U|}$   
    Compute  $\rho_a \leftarrow 1 - NE_a / \log_2 |U|$   
    Compute  $\varepsilon_a \leftarrow \sigma_a / (1 + \rho_a)$   
**end for**  
**return**  $\{\varepsilon_a\}_{a \in A_{num}}$

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The distance  $\Delta_A$  is bounded in  $[0, \sqrt{|A|}]$ , where 0 indicates that two objects are identical or highly similar, and higher values indicate greater dissimilarity. To ensure all values lie in  $[0, 1]$ , all numerical attributes are first normalized via min-max normalization.

Instead of relying on a single global radius for all attributes, we compute an adaptive radius for each numerical attribute based on its standard deviation and an entropy-based density factor.

**Definition 2.** For a numerical attribute  $a$ , let the objects in  $U$  be sorted in ascending order such that  $x_1^a \leq x_2^a \leq \dots \leq x_n^a$ . The gaps between adjacent values are  $\Delta_i^a = x_{i+1}^a - x_i^a$  for  $i = 1, \dots, n-1$ . Using the standard deviation  $\sigma_a$  as a reference, a connected component splits when  $\Delta_i^a > \sigma_a$ . The attribute neighborhood entropy is defined as:

$$NE_a = -\sum_{i=1}^{|C|} \frac{|C_i|}{|U|} \log_2 \frac{|C_i|}{|U|} \quad (8)$$

where  $\{C_1, \dots, C_{|C|}\}$  are the connected components formed by splitting at gaps exceeding  $\sigma_a$ . This entropy measures the uniformity of the attribute's value distribution.

Based on this entropy, the density factor  $\rho_a$  is computed as:

$$\rho_a = 1 - \frac{NE_a}{\log_2 |U|} \quad (9)$$

which ranges from 0 to 1. The adaptive radius  $\varepsilon_a$  for attribute  $a$  is then:

$$\varepsilon_a = \frac{\sigma_a}{1 + \rho_a} \quad (10)$$

where  $\sigma_a$  is the standard deviation of attribute  $a$ . In high-density regions ( $\rho_a \rightarrow 1$ ), the radius shrinks by up to 50%; in sparse regions ( $\rho_a \rightarrow 0$ ), it remains at  $\sigma_a$ .

The overall procedure is summarized in Algorithm 1.

### B. NMI-Based Graph Construction

In anomaly detection, whether an object is abnormal depends not only on its individual attribute values, but also on its relational context with other objects. An object that deviates from the local structural patterns of its neighbors is more likely to be anomalous.

A natural intuition is that two objects should be considered structurally similar if their neighborhoods share common

elements. This idea is analogous to the concept of mutual information in information theory and leads to the definition of Neighborhood Mutual Information (NMI).

**Definition 3.** Given the neighborhood knowledge  $n_a^{\varepsilon_a}(x)$  of object  $x$  with respect to attribute  $a$ , the object neighborhood entropy of  $x$  is defined as:

$$NE(x) = -\sum_{a \in A} \frac{|n_a^{\varepsilon_a}(x)|}{|U|} \log_2 \frac{|n_a^{\varepsilon_a}(x)|}{|U|} \quad (11)$$

**Definition 4.** For objects  $x, y \in U$ , the neighborhood joint entropy with respect to attribute set  $A$  is defined as:

$$NJE(x, y) = -\sum_{a \in A} \frac{|n_a^{\varepsilon_a}(x) \cap n_a^{\varepsilon_a}(y)|}{|U|} \log_2 \frac{|n_a^{\varepsilon_a}(x) \cap n_a^{\varepsilon_a}(y)|}{|U|} \quad (12)$$

**Definition 5.** The neighborhood mutual information between two objects  $x, y \in U$  with respect to attribute set  $A$  is defined as:

$$NMI(x, y) = NE(x) + NE(y) - NJE(x, y) \quad (13)$$

The expanded form is:

$$NMI(x, y) = \sum_{a \in A} \frac{|n_a^{\varepsilon_a}(x) \cap n_a^{\varepsilon_a}(y)|}{|U|} \log_2 \frac{|U| \cdot |n_a^{\varepsilon_a}(x) \cap n_a^{\varepsilon_a}(y)|}{|n_a^{\varepsilon_a}(x)| \cdot |n_a^{\varepsilon_a}(y)|} \quad (14)$$

with the convention that  $0 \log_2 0 = 0$ .

The NMI measure satisfies the following properties:

**Non-negativity.**  $NMI(x, y) \geq 0$ , with equality if and only if  $n_a^{\varepsilon_a}(x) \cap n_a^{\varepsilon_a}(y) = \emptyset$  for all  $a \in A$ , i.e., when the neighborhood structures of  $x$  and  $y$  have no overlap across any attribute.

**Symmetry.**  $NMI(x, y) = NMI(y, x)$ , which follows directly from the symmetric construction and ensures that the resulting graph is undirected.

**Structural discriminability.** A larger NMI value indicates stronger structural similarity between the two objects, suggesting that they reside in comparable local contexts across multiple attribute views. In contrast, a small NMI implies that the two objects have distinct neighborhood structures and are likely from different regions of the data manifold.

Based on NMI, a weighted undirected graph  $\mathcal{G} = (V, \mathcal{E})$  can be constructed, where each object  $x_i \in U$  serves as a node. The weight  $\mathbf{W}_{ij}$  of the edge between nodes  $x_i$  and  $x_j$  is defined as:

$$\mathbf{W}_{ij} = \frac{NMI(x_i, x_j)}{\sqrt{NE(x_i) \cdot NE(x_j)}} \quad (15)$$

Since  $NMI(x, y) \leq \min(NE(x), NE(y))$ , we have  $\mathbf{W}_{ij} \in [0, 1]$ , with  $\mathbf{W}_{ij} = 1$  when  $x_i$  and  $x_j$  share identical neighborhood structures.

To reduce graph density and suppress noisy connections, a threshold-based sparsification strategy is applied. The entries of the resulting sparse adjacency matrix are:

$$\mathbf{A}_{ij} = \begin{cases} \mathbf{W}_{ij}, & \text{if } \mathbf{W}_{ij} \geq \tau \\ 0, & \text{otherwise} \end{cases} \quad (16)$$

where  $\tau$  is a pre-defined threshold. The NMI-based graph construction process is detailed in Algorithm 2.

**Algorithm 2** NMI-Based Graph Construction

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**Input:** Data  $X$ , adaptive radii  $\{\varepsilon_a\}$ , threshold  $\tau$   
**Output:** Sparse adjacency matrix  $A$   
**for each**  $a \in A$  **do**  
    Compute neighborhood  $n_a^{\varepsilon_a}(x_i)$  for all  $x_i \in U$   
**end for**  
**for each**  $x_i \in U$  **do**  
    Compute  $NE(x_i)$  by Eq. (12)  
**end for**  
**for each pair**  $(x_i, x_j), i < j$  **do**  
    Compute  $NMI(x_i, x_j)$  by Eq. (14)  
    Compute  $W_{ij}$  by Eq. (16)  
    **if**  $W_{ij} \geq \tau$  **then**  
         $A_{ij} \leftarrow W_{ij}, A_{ji} \leftarrow W_{ij}$   
    **end if**  
**end for**  
**return**  $A$

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TABLE I  
THE INFORMATION SYSTEM  $\mathcal{I}$

| Object | $a_1$ | $a_2$ | $a_3$ |
|--------|-------|-------|-------|
| $x_1$  | 0.2   | 0.8   | A     |
| $x_2$  | 0.3   | 0.7   | A     |
| $x_3$  | 0.9   | 0.1   | B     |
| $x_4$  | 0.8   | 0.2   | B     |
| $x_5$  | 0.5   | 0.5   | A     |

**C. GCN-Based Anomaly Detection**

With the sparse graph  $\mathcal{G}$  constructed in the previous stage, anomaly detection is performed using a two-layer GCN in a semi-supervised manner. The GCN propagation rule and masked cross-entropy loss follow Eqs. (3)–(5) in Section 2.2. A small fraction of nodes (labeled ratio  $\alpha = 0.2$ ) are randomly selected for training, while the remaining nodes serve as the evaluation set.

After training, the anomaly score of each node  $x_i$  is defined as the predicted probability of belonging to the anomalous class:

$$s(x_i) = \mathbf{Z}_{i,1} \quad (17)$$

where  $\mathbf{Z} \in \mathbb{R}^{n \times 2}$  is the GCN output, and  $\mathbf{Z}_{i,1}$  is the probability for the anomalous class. A higher score indicates a greater likelihood of being an anomaly.

To determine the final set of outliers, an automatic threshold selection strategy based on the F1-score is adopted. The nodes are first sorted in descending order of their anomaly scores, and candidate thresholds are evaluated on the validation set. The threshold that maximizes the F1-score is selected as the optimal decision boundary.

**D. Illustrative Example**

To illustrate the proposed framework step by step, this subsection provides an example with the information system  $\mathcal{I}$  in Table I, where  $U = \{x_1, x_2, x_3, x_4, x_5\}$ ,  $A = \{a_1, a_2, a_3\}$ , with  $a_1, a_2$  being numerical and  $a_3$  categorical.

**Step 1: Adaptive radii.** Compute the mean and standard deviation of numerical attributes:  $\mu_{a_1} = 0.54$ ,  $\sigma_{a_1} = 0.291$ ;

TABLE II  
THE NEIGHBORHOOD KNOWLEDGE OF  $\mathcal{I}$

| Object | $n_{a_1}^{\varepsilon_{a_1}}$ | $n_{a_2}^{\varepsilon_{a_2}}$ |
|--------|-------------------------------|-------------------------------|
| $x_1$  | $\{x_1, x_2, x_5\}$           | $\{x_1, x_2\}$                |
| $x_2$  | $\{x_1, x_2, x_5\}$           | $\{x_1, x_2\}$                |
| $x_3$  | $\{x_3, x_4\}$                | $\{x_3, x_4\}$                |
| $x_4$  | $\{x_3, x_4\}$                | $\{x_3, x_4\}$                |
| $x_5$  | $\{x_1, x_2, x_5\}$           | $\{x_5\}$                     |

TABLE III  
NEIGHBORHOOD MUTUAL INFORMATION

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ |
|-------|-------|-------|-------|-------|-------|
| $x_1$ | 0.557 | 0.557 | 0.000 | 0.000 | 0.371 |
| $x_2$ | –     | 0.557 | 0.000 | 0.000 | 0.371 |
| $x_3$ | –     | –     | 0.464 | 0.464 | 0.000 |
| $x_4$ | –     | –     | –     | 0.464 | 0.000 |
| $x_5$ | –     | –     | –     | –     | 0.515 |

TABLE IV  
NORMALIZED NEIGHBORHOOD MUTUAL INFORMATION (EDGE WEIGHTS)

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ |
|-------|-------|-------|-------|-------|-------|
| $x_1$ | 1.000 | 1.000 | 0.000 | 0.000 | 0.694 |
| $x_2$ | –     | 1.000 | 0.000 | 0.000 | 0.694 |
| $x_3$ | –     | –     | 1.000 | 1.000 | 0.000 |
| $x_4$ | –     | –     | –     | 1.000 | 0.000 |
| $x_5$ | –     | –     | –     | –     | 1.000 |

$\mu_{a_2} = 0.46$ ,  $\sigma_{a_2} = 0.291$ . Taking attribute  $a_1$  as an example, the sorted values are  $\{0.2, 0.3, 0.5, 0.8, 0.9\}$ . The gaps are  $\{0.1, 0.2, 0.3, 0.1\}$ . With threshold  $\sigma_{a_1} = 0.291$ , the sequence splits at gap 0.3, yielding two connected components:  $C_1 = \{x_1, x_2, x_5\}$  and  $C_2 = \{x_4, x_3\}$ . By Definition 2,  $NE_{a_1} = -(\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5}) = 0.971$ .

Similarly,  $NE_{a_2} = 0.971$ , and for categorical attribute  $a_3$ ,  $NE_{a_3}$  is computed from its value distribution. The adaptive radii are then  $\varepsilon_{a_1} = 0.175$ ,  $\varepsilon_{a_2} = 0.175$ .

**Step 2: Neighborhood knowledge.** By Definition 1 and the adaptive radii, the neighborhood knowledge is obtained and summarized in Table II.

**Step 3: NMI computation.** The neighborhood information entropy for each object is computed:  $NE(x_1) = 0.557$ ,  $NE(x_2) = 0.557$ ,  $NE(x_3) = 0.464$ ,  $NE(x_4) = 0.464$ ,  $NE(x_5) = 0.515$ . By Definition 5, the NMI between all pairs is computed and shown in Table III.

**Step 4: Graph construction.** Edge weights are computed by Eq. (16). The resulting weight matrix is shown in Table IV, and the corresponding graph  $\mathcal{G}$  is illustrated in Fig. 2.

**IV. EXPERIMENTS****A. Data Preprocessing**

[To be completed: description of the 30 UCI datasets, min-max normalization, one-hot encoding, outlier labeling strategy, and dataset statistics (Table 5).]

**B. Comparison Algorithms and Parameter Settings**

[To be completed: descriptions of ADFNR, DASOD, GCN, GCN-LOF, NIEOD, and NMIGOD. Parameter settings and

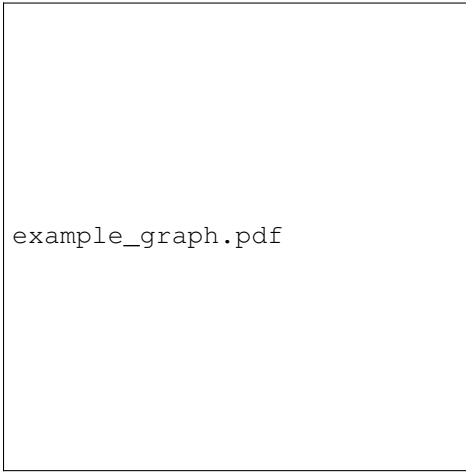


Fig. 2. The constructed NMI graph  $\mathcal{G}$  for the illustrative example.

grid search results (Table 6). Evaluation metrics: Precision, Recall, F1-score, and AUC.]

### C. Experimental Results and Analysis

1) *Overall Performance*: [To be completed: Precision, Recall, F1-score, and AUC comparison tables (Tables 7–10). Analysis of NMIGOD’s performance advantages across the 30 datasets. Win count summary and discussion of cases where NMIGOD excels or underperforms.]

2) *ROC and F1 Curve Analysis*: [To be completed: ROC curves for all algorithms on 30 datasets (Fig. 5). Average F1 curve (Fig. 6) and average ROC curve (Fig. 7). Discussion of ranking capability and threshold-independent performance.]

3) *Ablation Study*: [To be completed: ablation experiments on four representative datasets (iris, wine, glass, diabetes). Comparison of full NMIGOD vs. NMIGOD without adaptive radius vs. NMIGOD with global radius. Analysis of the contribution of the adaptive radius mechanism and NMI graph construction.]

4) *Statistical Significance Tests*: [To be completed: Friedman test and Nemenyi post-hoc test results. Analysis of statistical significance of NMIGOD’s improvements over comparison methods.]

### D. Parameter Sensitivity Analysis

[To be completed: analysis of the effect of  $\tau$  (MI threshold),  $\lambda$  (radius coefficient), GCN hidden dimensions, and labeled ratio on detection performance.]

### E. Discussion

[To be completed: summary of findings, limitations of NMIGOD, analysis of failure cases (e.g., yeast, hepatitis, arrhythmia), and directions for future work.]

## V. CONCLUSION

## ACKNOWLEDGMENTS

## REFERENCES

[1] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly detection: A survey,” *ACM Comput. Surv.*, vol. 41, no. 3, pp. 1–58, 2009.

[2] C. C. Aggarwal, *Outlier Analysis*, 2nd ed. Springer, 2017.

[3] E. W. Ngai, Y. Hu, Y. H. Wong, Y. Chen, and X. Sun, “The application of data mining techniques in financial fraud detection,” *Decis. Support Syst.*, vol. 50, no. 3, pp. 559–569, 2011.

[4] A. L. Buczak and E. Guven, “A survey of data mining and machine learning methods for cyber security intrusion detection,” *IEEE Commun. Surveys Tuts.*, vol. 18, no. 2, pp. 1153–1176, 2016.

[5] P. J. Rousseeuw and A. M. Leroy, *Robust Regression and Outlier Detection*. John Wiley & Sons, 1987.

[6] P. Cunningham and S. J. Delany, “k-Nearest neighbour classifiers (with Python examples),” *arXiv preprint arXiv:2004.04523*, 2020.

[7] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, “LOF: Identifying density-based local outliers,” in *Proc. ACM SIGMOD*, 2000, pp. 93–104.

[8] Z. He, X. Xu, and S. Deng, “Discovering cluster-based local outliers,” *Pattern Recognit. Lett.*, vol. 24, no. 9–10, pp. 1641–1650, 2003.

[9] G. Pang, L. Cao, L. Chen, and H. Liu, “Unsupervised feature selection for outlier detection by modelling hierarchical value-feature couplings,” in *Proc. ICDM*, 2016, pp. 410–419.

[10] J. He, Q. Xu, Y. Jiang, Z. Wang, and Q. Huang, “ADA-GAD: Anomaly-denoised autoencoders for graph anomaly detection,” in *Proc. AAAI*, vol. 38, 2024, pp. 8481–8489.

[11] Z. Pawlak, “Rough sets,” *Int. J. Comput. Inf. Sci.*, vol. 11, no. 5, pp. 341–356, 1982.

[12] T. Y. Lin, “Neighborhood systems and relational databases,” in *Proc. 16th ACM Annu. Conf. Comput. Sci.*, 1988, p. 725.

[13] Q. H. Hu, D. R. Yu, J. F. Liu, and C. X. Wu, “Neighborhood rough set based heterogeneous feature subset selection,” *Inf. Sci.*, vol. 178, no. 18, pp. 3577–3594, 2008.

[14] Z. Yuan, X. Zhang, and S. Feng, “Hybrid data-driven outlier detection based on neighborhood information entropy and its developmental measures,” *Expert Syst. Appl.*, vol. 112, pp. 243–257, 2018.

[15] D. R. Wilson and T. R. Martinez, “Improved heterogeneous distance functions,” *J. Artif. Intell. Res.*, vol. 6, no. 1, pp. 1–34, 1997.

[16] Z. Yuan, H. Chen, T. Li, J. Liu, and S. Wang, “Anomaly detection based on fuzzy neighborhood rough sets,” *Inf. Sci.*, vol. 709, p. 122075, 2025.

[17] R. Wille, “Restructuring lattice theory: An approach based on hierarchies of concepts,” in *Ordered Sets*. Springer, 1982.

[18] B. Ganter and R. Wille, *Formal Concept Analysis: Mathematical Foundations*. Springer, 2012.

[19] Q. Hu, K. Qin, L. Zhang, and J. Li, “A novel outlier detection approach based on formal concept analysis,” *Knowl.-Based Syst.*, vol. 268, p. 110486, 2023.

[20] J. Li, K. Qin, L. Zhang, and Q. Hu, “Dual-aspect synergistic outlier detection with structural deviation and attribute rarity,” *Pattern Recognit.*, vol. 180, p. 114084, 2026.

[21] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *Proc. ICLR*, 2017.

[22] A. Goode, B. Hooi, S. K. Ng, and W. S. Ng, “LUNAR: Unifying local outlier detection methods via graph neural networks,” in *Proc. AAAI*, vol. 36, no. 6, 2022, pp. 6737–6745.

[23] K. Ding, J. Li, R. Bhanushali, and H. Liu, “Deep anomaly detection on attributed networks,” in *Proc. SDM*, 2019, pp. 594–602.

[24] X. Du, J. Li, Z. Chen, and Y. Wang, “Graph autoencoder-based unsupervised outlier detection,” *Inf. Sci.*, vol. 608, pp. 532–550, 2022.

[25] Q. Li, Z. Han, and X.-M. Wu, “Deeper insights into graph convolutional networks for semi-supervised learning,” in *Proc. AAAI*, vol. 32, no. 1, 2018.

[26] K. Liu, Y. Dou, Y. Zhao, X. Ding, X. Hu, R. Zhang, K. Ding, C. Chen, H. Peng, K. Shu, *et al.*, “Graph-based anomaly detection and deep learning: A comprehensive survey,” *ACM Comput. Surv.*, vol. 55, no. 8, pp. 1–37, 2022.

[27] F. Jiang, Y. Sui, and C. Cao, “An information entropy-based approach to outlier detection in rough sets,” *Expert Syst. Appl.*, vol. 37, no. 9, pp. 6338–6344, 2010.

[28] G. Pang, C. Shen, L. Cao, and A. V. D. Hengel, “Deep learning for anomaly detection: A review,” *ACM Comput. Surv.*, vol. 54, no. 2, pp. 1–38, 2021.

[29] G. Duan, H. Lv, H. Wang, and G. Feng, “Application of a dynamic line graph neural network for intrusion detection,” *IEEE Trans. Inf. Forensics Secur.*, vol. 18, pp. 699–714, 2023.

[30] H. Yuan, H. Yu, S. Gui, and S. Ji, “Explainability in graph neural networks: A taxonomic survey,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 6352–6372, 2022.

[31] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Proc. NeurIPS*, vol. 30, 2017.

[32] W. Hamilton, Z. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Proc. NeurIPS*, vol. 30, 2017.

- [33] M. Feurer and F. Hutter, "Hyperparameter optimization," in *Automated Machine Learning*, Springer, 2019, pp. 3–33.
- [34] C. Z. Wang, Q. H. Hu, X. Z. Wang, D. G. Chen, and Y. H. Qian, "Feature subset selection based on fuzzy neighborhood rough sets," *Knowl.-Based Syst.*, vol. 111, pp. 173–179, 2016.
- [35] X. Su, J. Zhou, Y. Chen, and W. Pedrycz, "Identifying outliers via local granular-ball density," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 18956–18967, 2025.
- [36] J. Yang, X. Yang, J. Zhou, and Y. Yao, "Three-way outlier detection based on shadowed granular-balls," *IEEE Trans. Fuzzy Syst.*, vol. 34, no. 1, pp. 101–113, 2026.
- [37] A. Roy, J. Li, and S. Pan, "GAD-NR: Graph anomaly detection via neighborhood reconstruction," in *Proc. WSDM*, 2024, pp. 576–585.
- [38] N. Rahimi and R. Javadi, "Application of graph autoencoder in outlier detection," *Appl. Soft Comput.*, 2026.
- [39] Z. Qin, L. Zhang, and K. Li, "Enhancing intrusion detection performance using GCN-LOF," *Comput. Netw.*, 2025.
- [40] Y. Y. Yao, "Relational interpretations of neighborhood operators," *Inf. Sci.*, vol. 111, no. 1–4, pp. 239–259, 1998.
- [41] Y. M. Chen, D. Q. Miao, and R. Z. Wang, "Neighborhood outlier detection," *Expert Syst. Appl.*, vol. 37, no. 12, pp. 8745–8749, 2010.
- [42] L. A. Zadeh, "Fuzzy sets and information granularity," in *Advances in Fuzzy Set Theory and Applications*, 1979, pp. 3–18.
- [43] D. Dubois and H. Prade, "Rough fuzzy sets and fuzzy rough sets," *Int. J. Gen. Syst.*, vol. 17, no. 2–3, pp. 191–209, 1990.
- [44] Z. Yuan, H. Chen, T. Li, B. Sang, and S. Wang, "Outlier detection based on fuzzy rough granules in mixed attribute data," *IEEE Trans. Cybern.*, vol. 52, no. 8, pp. 8399–8412, 2021.
- [45] Z. Li, Y. Zhao, J. Han, Y. Su, R. Jiao, and X. Wen, "A survey on explainable anomaly detection," *ACM Trans. Knowl. Discov. Data*, vol. 18, no. 1, pp. 1–54, 2024.
- [46] N. J. Mlyahilu and E. Novikova, "Graph construction approaches for graph neural networks," preprint, 2025.
- [47] F. G. Febrinanto, K. Moore, C. Thapa, M. Liu, V. Saikrishna, J. Ma, and F. Xia, "Entropy causal graphs for multivariate time series anomaly detection," *ACM Trans. Inf. Syst.*, 2025.
- [48] S. He, G. Li, K. Xie, and P. K. Sharma, "Fusion graph structure learning-based multivariate time series anomaly detection with structured prior knowledge," *IEEE Trans. Inf. Forensics Secur.*, 2024.